

## Reconstruction of Annual Temperature (1590–1979) for Longmire, Washington, Derived from Tree Rings

LISA J. GRAUMLICH<sup>1</sup> AND LINDA B. BRUBAKER

*College of Forest Resources, AR-10, University of Washington, Seattle, Washington 98195*

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Annual growth records from trees at timberline in the Cascade Range of Washington are correlated with variations in temperature and snow depth and used to reconstruct climatic variation in the past. Response surfaces indicate that growth of mountain hemlock (*Tsuga mertensiana*) and subalpine larch (*Larix lyallii*) is positively correlated with summer (July to September) temperature and negatively correlated with spring (March) snow depth when snow depth is at or below average. During years of above average snow depth, temperature has little effect on mountain hemlock but has a negative effect on growth in subalpine larch. These interactions make it difficult to reconstruct these climatic variables separately using standard methods. Mean annual temperature values, which combine information on both summer temperature and spring snow depth, were estimated from a regression model that reconstructs past temperature at Longmire, Washington, as a function of larch and hemlock tree-ring chronologies. The reconstruction of mean annual temperature shows temperatures between 1590 and 1900 to be approximately 1°C lower than those of the 20th century. Only during a short period from 1650 to 1690 did temperatures approach 20th-century values. © 1986 University of Washington.

### INTRODUCTION

Regional variation in the magnitude of Little Ice Age (c. 1300 to 1850 A.D.; Lamb, 1982) temperature depression and in the timing of subsequent warming to modern conditions is well documented (Jones and Kelly, 1983; Fritts and Lough, 1985). In the Pacific Northwest proxy temperature data, such as records of tree growth and times of glacial moraine formation, indicate that temperatures during the Little Ice Age were lower than those experienced in the 20th century. The magnitude of this cooling is uncertain: temperature reconstructions based on trees in nearby semi-arid regions indicate that the cooling in the Pacific Northwest was slight (0.3°C) (Fritts *et al.*, 1979), while the position and timing of moraine formation in the Cascade Range of Washington imply that average annual temperatures were 1°C lower during the Little Ice Age (Burbank, 1981, 1982).

A previously untapped source of information on past temperatures in the Pacific Northwest is the annual ring-width variation of trees at high elevations where temperature limits annual growth. The purpose of this study is (1) to describe the growth response of subalpine trees in the Washington Cascade Range to variations in temperature and snow depth, and (2) to reconstruct temperature variation over the last 400 yr.

The effect of climate on the growth of subalpine trees can be characterized by response surfaces (Bartlein *et al.*, 1986) that portray the simultaneous effect of two or more climatic variables on tree growth. Response surfaces provide a means of determining the form of tree growth response to individual climatic variables (i.e., linear, exponential, sigmoidal, etc.) and of detecting the presence of interactions between the climatic variables as they influence growth. Interactions between climatic variables in controlling growth are suspected when the response of growth to one variable changes depending on the value of

<sup>1</sup> Present address: Department of Ecology and Behavioral Biology, University of Minnesota, Minneapolis, Minnesota 55455.

a second variable. The results of the response surface analysis are used here (1) to reveal possible indeterminacies in the relationship between tree growth and climate, (2) to determine the form of the model used for reconstructing past climate, and (3) to refine the interpretation of the final reconstruction.

A multiple regression model is then used to reconstruct climate based on the tree-ring data. The general procedures for reconstructing climate from tree-ring data are well documented (Fritts, 1976) as are the biological and statistical assumptions underlying the use of multiple regression to reconstruct climate (Lofgren and Hunt, 1982). In this paper the standard dendroclimatic procedures are complemented with methods designed to uncover violations of the assumptions underlying multiple regression, and to identify values of the reconstructed series that are extrapolations. The temperature reconstruction for Longmire is then compared with other records of climatic variation in the western United States to assess its correspondence to both local and regional climatic records.

### DATA

Temperature, precipitation, and snow data were compiled from three weather stations located at high elevations in the Cascade Range (Fig. 1). The temperature and precipitation record at Longmire (46°47'N, 121°44'W; 842 m) extends back to 1916 and is the longest climatic record available from a high-elevation station (>750 m) in Washington state. Monthly records of snow depth from the Paradise Ranger Station (46°47', 121°44'; 1694 m) extend back to 1917 with 3 yr of missing data. Snow data at Stevens Pass (47°45', 121°05'; 1245 m) extend back to 1939 with 6 yr of missing data.

Ring-width chronologies were derived from increment cores collected from four stands of mountain hemlock (*Tsuga mertensiana*), and three stands of subalpine larch (*Larix lyallii*) growing at or near timberline (1700 to 2400 m; Graumlich, 1985;

Fig. 1). Two cores were collected per tree, 22 to 30 trees were sampled at each site, and all specimens were cross-dated to insure that the correct calendar year was assigned to each ring-width measurement. Standardization of individual ring-width series was accomplished by fitting a negative exponential or horizontal curve to each series thereby removing age-associated trends in growth (Fritts, 1976).

### GROWTH RESPONSE OF SUBALPINE TREES TO CLIMATE

The response of trees to climate must be understood before past climatic variation can be inferred from patterns of tree growth. Dendroclimatologists traditionally have used multiple regression equations that express tree growth as a function of the eigenvalues of monthly temperature and precipitation for this purpose (Fritts, 1976). The partial regression coefficients associated with each month's temperature or precipitation series form what has been termed "response functions" which are interpreted as indicating the direction and magnitude of the response of tree growth to each month's climate. These traditional response functions describe the linear, additive effect of each month's climate on tree growth, and as such, are poorly suited for revealing nonlinear relationships between climate and tree growth, or interactions among the climatic variables.

Response surfaces, in which tree growth is expressed as a polynomial function of two or more climatic variables, provide an alternative method of displaying the relationship between tree growth and climate. The response surface equations will typically include the squares and cross-products of the climatic variables, and as such they are able to represent nonlinear relationships that include interactions among the climatic variables. In this section the effect of growing season temperature and depth of spring snow pack on tree growth are investigated using response surface methods.

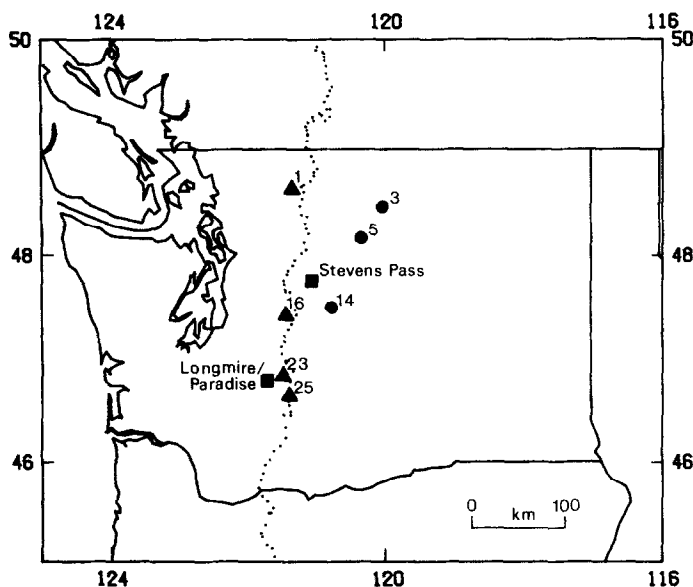


FIG. 1. Map of Washington showing locations of subalpine larch sites (circles), mountain hemlock sites (triangles), and climatic stations (squares). Crest of the Cascade Range is indicated by the dotted line.

### Methods

For each site, bivariate scatter plots of tree growth versus summer (July to September) temperature, summer precipitation, and winter (January) and spring (March) snow depth demonstrated the strength of the correlation between these climatic variables and tree growth. Because of their geographic proximity, snow data from Paradise were used in the analysis of the mountain hemlock chronologies while snow data from Stevens Pass were used with the subalpine larch chronologies. The climatic variables most strongly related to the growth of both mountain hemlock and subalpine larch are summer temperature and spring snow depth (Graumlich, 1985). Response functions indicated a positive response to temperature from April to July but no consistently significant response to monthly precipitation.

Response surfaces display tree growth as a surface, the height of which represents the magnitude of the growth response to a given combination of two climatic variables (Bartlein *et al.*, 1986). To estimate the response surface for each species,  $x$  and  $y$

axes were constructed that define a plane representing all possible combinations of summer temperature (the  $x$  variable) and spring snow depth (the  $y$  variable). For each species, yearly values of tree growth averaged across sites were plotted on a third ( $z$ ) axis at the point on the  $x$ - $y$  plane representing the climatic conditions of that year. A major assumption of this approach, that the  $x$ - $y$  plane is covered by observations, is met because summer temperature and spring snow depth are uncorrelated. The individual observations were smoothed to form a more generalized surface by fitting second-degree polynomial equations to the data using multiple regression techniques (Mead and Pike, 1975). A second-degree polynomial equation was used to allow for interaction between climatic variables and nonlinearity in tree response.

### *Response Surfaces for Mountain Hemlock and Subalpine Larch*

Response-surface equations for spring snow depth and summer temperature explain 35% of the variance of tree growth for mountain hemlock and 39% for subalpine

larch and reveal nonlinear relationships between tree growth and climate variables (Table 1; Figs. 2a, b). The response surfaces for both species are highest, indicating greatest tree growth, when summers are warm and snow depth is minimal. When snow depths are average or less, both species show simple patterns of increasing growth with increasing temperature. The height of the response surface for mountain hemlock is not appreciably affected by changing values of temperature when spring snow depth reaches values of 2 m or more, indicating that temperature has little effect on growth under these conditions. The response surface for subalpine larch, in contrast, indicates that growth is negatively affected by summer temperature when snow depth is above average.

The positive response of these species to increasing temperature at low to moderate snow depths undoubtedly reflects the positive influence of temperature on physiological processes related to growth (Tranquillini, 1979). When deep snow persists into the late spring and early summer, however, soil temperatures remain cold due to the insulation of snow and the ability of the trees to absorb water from the soil is severely reduced. Physiological studies of other subalpine tree species have shown that, at low soil temperatures, water movement into the tree decreases sharply resulting in stomatal closure and decreased carbon assimilation (Teskey *et al.*, 1984; Tranquillini, 1979). The effect of deep snow on mountain hem-

lock therefore is to dramatically decrease growth regardless of summer temperature, whereas the effect on subalpine larch is most pronounced at warm summer temperatures (Fig. 2). A combination of cold soil temperatures and warm air temperatures would increase the evaporative demands on needles and lead to a reduction in growth due to dessication damage (Tranquillini, 1979). The deciduous needles of larch probably dessicate more rapidly than the evergreen mountain hemlock needles because the relatively thin cuticle of larch needles cannot as efficiently impede transpiration loss under conditions of cold soils and reduced water uptake.

*Implications For Climatic Reconstruction*

Despite the existence of a significant correlation between summer temperature and tree growth, a reconstruction derived from a regression model that expresses summer temperature as a function of tree growth will be unreliable because the model cannot account for the effect of variation in snow depth on tree growth. A climatic variable that reflects the simultaneous variation in both summer temperature and spring snow depth will be more readily reconstructed by the available tree-ring data. Mean annual temperature reflects the intensity of summer warmth as well as the severity of winter conditions that govern the total depth of snow accumulated by spring. For example, the average temperature at Longmire for the

TABLE 1. RESPONSE SURFACE REGRESSION RESULTS

| Species          | <i>b</i> <sub>0</sub> | <i>b</i> <sub>10</sub> | <i>b</i> <sub>01</sub> | <i>b</i> <sub>20</sub> | <i>b</i> <sub>02</sub> | <i>b</i> <sub>11</sub> | <i>R</i> <sup>2a</sup> | <i>R</i> <sup>2<sub>a</sub>b</sup> |
|------------------|-----------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------------------|
| Mountain Hemlock | 0.421                 | -0.174                 | 0.652                  | 0.731                  | -0.926                 | -0.374                 | 34.8                   | 28.0                               |
| Subalpine Larch  | -0.015                | 1.644                  | 0.491                  | -0.482                 | -0.146                 | -1.813                 | 38.8                   | 27.4                               |

*Note.* The equation used to predict tree growth as a function of temperature and snow depth is

$$Z = b_0 + b_{10}X + b_{01}Y + b_{20}X^2 + b_{02}Y^2 + b_{11}XY$$

where

*Z* = tree-growth indices averaged across sites for each species; *X* = summer (July to September) average temperature (in °C); *Y* = spring (March) snow depth (in m).

<sup>a</sup> *R*<sup>2</sup> = squared multiple correlation coefficient of the regression equation.

<sup>b</sup> *R*<sup>2<sub>a</sub></sup> = *R*<sup>2</sup> adjusted for loss of degrees of freedom based on the number of predictors in the equation.

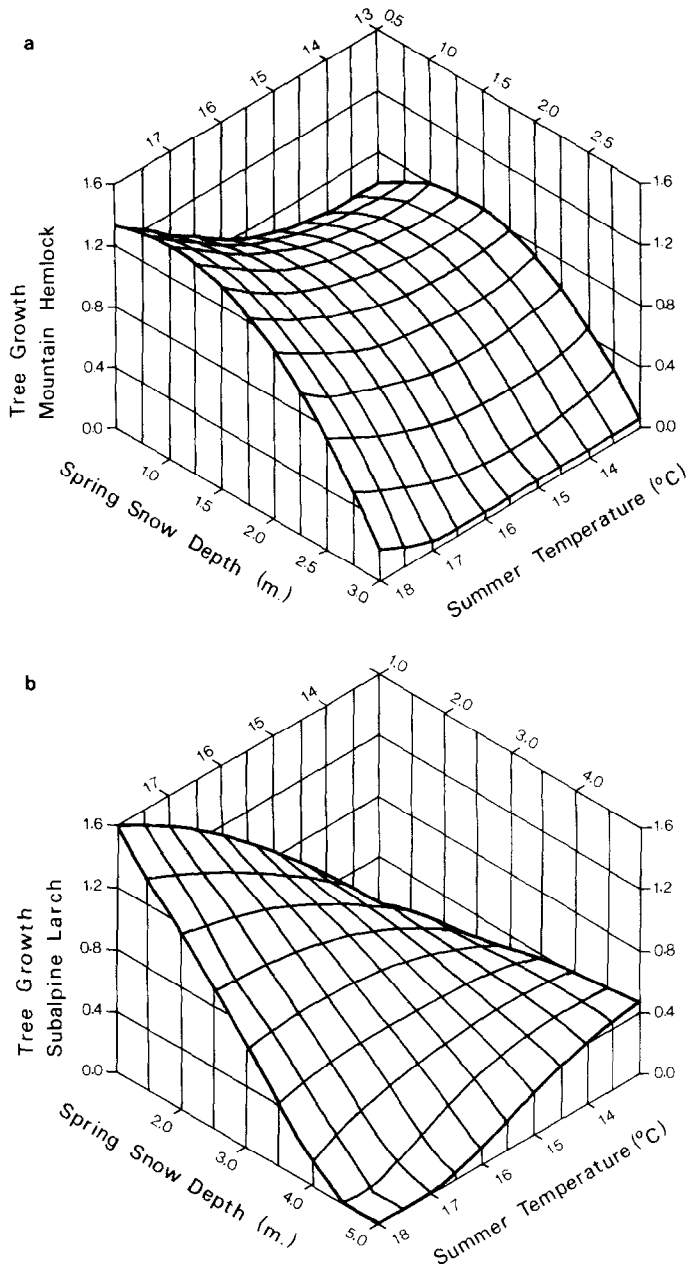


FIG. 2. Response surfaces for (a) mountain hemlock and (b) subalpine larch. Height of each surface represents the magnitude of growth response to a given combination of summer (July to September) temperature and spring (March) snow depth conditions. Tree growth indices are dimensionless units with mean equal to 1.0 and represent growth averaged across sites for each species.

period November to March is significantly and negatively correlated with March snow depth measured at Paradise ( $r = -0.54$ ,  $n = 63$ ). A reconstruction of mean annual temperature would thus implicitly incorpo-

rate the combined effect of both summer temperature and spring snow depth on growth. Information on climatic variation at the seasonal time scale, however, can only be gained by qualitative inferences

based on contrasting patterns of growth between species with differing climatic responses (see Discussion). Response surfaces have the potential of generating quantitative estimates of past seasonal climates by exploiting variation in the shape of the response surfaces among species (Prentice, 1984).

### RECONSTRUCTION OF MEAN ANNUAL TEMPERATURE

Quantitative reconstructions of climate based on tree-ring data are derived from regression models in which the tree-ring chronologies are the predictor (independent) variables and the climatic series is the predicted (dependent) variable. Regression equations are easily derived using standard procedures implemented on widely available software. Insuring the statistical reliability of a climatic reconstruction is more difficult. Reliability can be evaluated by the application of (1) regression diagnostic procedures to assess whether the assumptions underlying regression analysis are violated by the model, (2) cross-validation or verification procedures to test the model against data withheld from the model-building process, and (3) statistics that indicate whether the model is extrapolating when predicting past temperature.

#### *Regression Procedures*

Best subset regression procedures (Draper and Smith, 1981) were used to derive a model for predicting mean annual temperature at Longmire. Best subset regression makes use of an efficient algorithm to examine the very large number of regression equations resulting from every possible combination of predictor variables. Candidate predictor variables included seven high-elevation tree-ring chronologies (Fig. 1) as well as these same chronologies lagged forward by one year. The tree-ring data were lagged forward because climate in a given year affects physiological processes that in turn affect diam-

eter growth in ensuing years (Fritts, 1976). The primary criterion for choosing predictor variables was minimization of Mallows's  $C_p$  statistic, a measure of the trade-off between the goodness-of-fit of the regression as described, for example, by the explained variance and the number of parameters in the model (Draper and Smith, 1981).

The statistical assumptions of multiple regression analysis that are most commonly violated in paleoclimatic models are that (1) the residuals (i.e., the differences between each predicted and observed value of the dependent variable) are independent, normally distributed, and do not exhibit any systematic changes in variance, (2) the predictor variables are not perfectly or very highly correlated with one another, and (3) the correct predictor variables have been included in the model and there is a linear relationship between the predictor variables and the dependent variable (Lofgren and Hunt, 1982). Violations of these assumptions can bias and decrease the precision of the estimates of the coefficients (Graumlich, 1985). Regression diagnostic procedures, which assess whether a model violates these statistical assumptions, were used to evaluate candidate models (Weisberg, 1980). The normality of the residuals for each model can be assessed by comparing normal probability plots of the residuals to representative plots (Weisberg, 1980, p. 134). The condition number of the predictor variable matrix (Weisberg, 1980) indicates the extent to which coefficients of the chosen models may be biased by the correlation between tree-ring variables. Lack of significant autocorrelation in the residuals and the lack of trend in plots of the residuals versus predicted values demonstrates that the residuals are independent.

Outliers, in this case observations that depart substantially from the trend of the majority of the data, can also bias the regression estimates and result in a regression model of reduced precision (Weisberg,

1980). Outliers may arise from short-term, nonclimatic influences on growth (i.e., fires, insect infestations) or from the influence of a climatic event not explicitly modeled (severe growing season frost, damage due to high winds). To minimize the influence of such anomalies, three observations with highly significant values of the Cook's distance statistic (Weisberg, 1980) were eliminated.

### Cross-Validation

In order to cross-validate or verify the model, one-third of the observations were randomly deleted and the best subset regression procedure was performed on the remaining data. The observations withheld from the analysis were compared to the model's predictions using the product-moment correlation coefficient ( $r$ ) and the reduction of error statistic (RE; Fritts, 1976). A value of RE greater than zero is taken to indicate that the reconstruction model provides a better prediction of past climate variation than a prediction based solely on the mean of the calibration data set (Fritts, 1976).

This model-building procedure was repeated on five different randomly chosen subsets of the data to insure that the results were representative. Models estimated with different subsets of observations varied only slightly in their ability to predict independent observations of temperature (Graumlich, 1985). The coefficients for the final model were reestimated using the entire 1914–1979 temperature series to ensure that the coefficient estimates were as precise as possible (Table 2). The final model includes two subalpine larch chronologies and two mountain hemlock chronologies. One of the larch and one of the hemlock chronologies are lagged forward by one year. Three of the four coefficients are positive, reflecting a positive correlation between growth and mean annual temperature. A negative coefficient for the forward lagged mountain hemlock chronology is not biologically meaningful and reflects

TABLE 2. REGRESSION RESULTS FOR MEAN ANNUAL TEMPERATURE RECONSTRUCTION

|                    |                 | Preliminary model | Final model |
|--------------------|-----------------|-------------------|-------------|
| Coefficients       | Site 5 (SL)     | 1.57*             | 1.28*       |
|                    | Site 14 (SL)    | 0.39              | 0.64        |
|                    | Site 16 (MH)    | 2.95              | 2.81        |
|                    | Site 25 (MH)    | -2.21*            | -2.19*      |
|                    | Intercept       | 41.06             | 41.18       |
| Summary statistics | $R^2_{adj}$     | 0.43              | 0.40        |
|                    | $R^2_{adj}$     | 0.36              | 0.36        |
|                    | AC <sup>c</sup> | 0.10              | 0.01        |
|                    | RE <sup>d</sup> | 0.38              | —           |
|                    | $r^e$           | 0.65              | —           |

Note. Site numbers correspond to those in Fig. 1. Species abbreviations: SL = subalpine larch; MH = mountain hemlock.

\* Chronology lagged forward by one year.

<sup>a</sup>  $R^2$  = squared multiple correlation coefficient of the regression equation.

<sup>b</sup>  $R^2_{adj}$  =  $R^2$  adjusted for loss of degrees of freedom based on the number of predictors in the equation.

<sup>c</sup> AC = first-order autocorrelation of residuals.

<sup>d</sup> RE = reduction of error statistic.

<sup>e</sup>  $r$  = correlation between independent observations and predictions made by the model.

the necessary adjustment of coefficient estimates when correlation exists among predictor variables (Draper and Smith, 1981).

### Determining If Predictions Are Extrapolations

When predictions are made from an empirically derived model, the model interpolates a predicted value when the new observations fall within the range of the original predictor data set. The model extrapolates, however, when new data fall outside the range of the original data. The range of chronology values over which it is reasonable to predict new values of climate can be approximated by the minimum covering ellipsoid (MCE) or the smallest volume ellipsoid that circumscribes the observed values (Weisberg, 1980; Graumlich, 1985). If observations of the tree-ring variables from before the period of recorded climate fall within the ellipse, the prediction arising from the model can be considered an interpolation; if the observations fall outside the ellipse, the prediction is an extrapolation. A prediction may be classified as an extrapolation even if it is based

on values of tree-ring variables that are within their individual ranges but in a combination not observed in the original data. An extrapolation is therefore not necessarily a prediction with an extremely high or low value.

### Results

The final model for reconstructing temperature as well as all of the candidate models passed the regression diagnostic tests (Table 2; Graumlich, 1985). Therefore, so far as possible, the impact of violations of the assumptions of regression analysis on the reliability of the results has been minimized.

The mean annual temperature reconstruction for Longmire shows distinct episodes of higher and lower temperatures (Fig. 3). The most striking feature of the reconstruction is the rise in temperatures commencing around 1900 and marking the end of the Little Ice Age. The mean reconstructed temperature (1590 to 1913) is 6.3°C, almost 1°C lower than the mean temperature of the observed record (7.2°C; 1914 to 1979). A short period of warmth almost equal to that of the 20th century occurred between 1650 and 1690. Cool episodes occurred from 1600 to 1650, from 1700 to 1760, and from 1860 to 1900.

Less than 10% of all the predictions were classed as extrapolations (Fig. 3). Many of the extrapolations are values close to the

mean value of the reconstruction and thus do not affect the long-term trends seen in the reconstruction. Extremely low values reconstructed for 1896, 1899, and 1913 and identified as extrapolations were checked against measured values of temperature from the longer climatic record at Seattle. In each case, temperatures far below the mean were observed in the instrumental record. The extrapolations are thus consistent with available observed data and the model can be considered valid when predicting low values that otherwise may be identified as extrapolations.

### DISCUSSION

#### *Longmire Temperature Record and Fluctuations of Cascade Mountain Glaciers*

The Longmire temperature reconstruction is corroborated by evidence that glaciers of the Cascade Mountains extended beyond their modern termini during the Little Ice Age (Miller, 1969; Burbank, 1981; Heikkinen, 1984). Burbank (1982) used the relationship between climatic variables and equilibrium line altitude for subalpine glaciers in the Cascade Range to estimate that a 1°C lowering of summer temperatures would be necessary for the documented extension of the Mount Rainier glaciers. Burbank (1981) found a high degree of synchrony in the timing of moraine stabilization for four glaciers on Mount Rainier

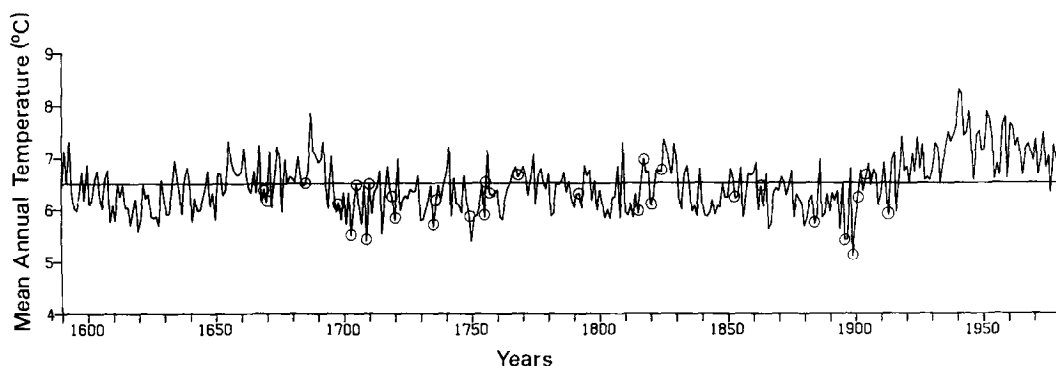


FIG. 3. Reconstruction of mean annual temperature for Longmire, Washington. Solid horizontal line is drawn through the mean of the reconstruction (1590 to 1913). Open circles indicate values that are statistically equivalent to extrapolations.



when he dated moraines using lichenometric, dendrochronologic, and tephrochronologic techniques. When times of moraine stabilization are compared to the Longmire temperature reconstruction (Fig. 4a), periods of moraine formation correspond to short-term trends from higher to lower temperatures (Fig. 4b). The only two exceptions to this pattern are the moraine building episodes from 1768 to 1777 and 1857 to 1863, both of which correspond to periods of average warmth. Moraine formation during these two periods could have been caused by changes in other variables governing mass balance such as snow accumulation, cloud cover, or humidity. The close correspondence between moraine stabilization and relatively short-term cooling trends implies that the lag between climatic events and the response of small glaciers is minimal.

#### Comparison with Other

##### *Dendroclimatic Reconstructions*

Cropper and Fritts (1985) reconstructed mean annual temperature for the Pasco River Basin in southeastern Washington based on large-scale patterns of tree growth at 65 semi-arid sites in the western United States and southwestern Canada. Whereas the temperature record for Longmire shows cool conditions throughout most of the 18th and 19th centuries followed by a 0.9°C rise in temperature beginning in the 20th century, the temperature record derived by Cropper and Fritts for the Pasco Basin shows little change in the mean over the last 300 yr (Fig. 4c). The absolute difference between these two estimates of temperature change at the beginning of the 20th century may be attributed to the differing sensitivities of subalpine versus semi-arid trees in responding to long-term changes in temperature.

Other long-term records of temperature variation for the western United States include a summer (July) degree day reconstruction for central Alaska and northwestern Canada (Fig. 4d; Jacoby *et al.*,

1985) and the Campito Mountain bristlecone pine (*Pinus longaeva*) chronology from California, which is strongly correlated with spring and summer temperatures (Fig. 4e; LaMarche and Stockton, 1974).

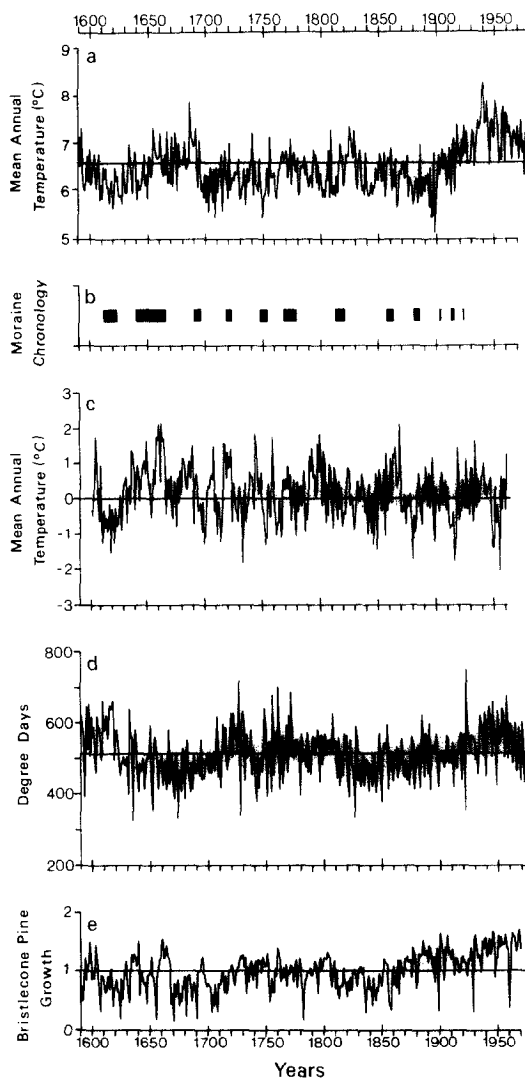


FIG. 4. Time series of proxy temperature data: (a) mean annual temperature reconstruction for Longmire, Washington, (b) periods of moraine stabilization at Mount Rainier, Washington (Burbank, 1981), (c) reconstruction of mean annual temperature for the Pasco Basin in southeastern Washington (Cropper and Fritts, 1985), (d) reconstruction of July degree days for central Alaska and northwestern Canada (Jacoby *et al.*, 1985), and (e) Campito Mountain bristlecone pine tree-ring chronology from the White Mountains of California (LaMarche and Stockton, 1974).

The most consistent similarity between these records and the Longmire temperature reconstruction is the warming trend at the end of the 19th century. During earlier centuries, the long-term variation in the Longmire reconstruction appears out of phase with the Alaskan record, most notably from 1600 to 1750.

Other than the late 19th century warming trend, the Longmire temperature reconstruction and the bristlecone pine chronology have no long-term patterns in common. V. C. LaMarche *et al.* (1984) have noted that growth of subalpine trees in California and Nevada has not decreased in response to the observed decrease in temperature that began about 1950, but instead has continued to rise. They speculated that these species are responding to carbon dioxide fertilization because the low partial pressure of carbon dioxide limits photosynthesis at the high elevations (3000 to 3500 m) where these trees grow. In contrast, growth of subalpine trees in Washington peaks in the 1940s and then declines, following regional temperature trends. Timberline in the Pacific Northwest is much lower (1400 to 2000 m) than in California and Nevada and subalpine trees appear to be unaffected by the postulated fertilizing effects of increased carbon dioxide.

#### *Seasonal Aspects of Temperature Variation at Longmire*

The response surfaces of mountain hemlock and subalpine larch together with their long-term growth histories can be used to characterize seasonal aspects of the temperature reconstruction. Above-average growth appears in the tree-ring records for both species for the periods 1920 to the present and 1650 to 1690 (Fig. 5) implying, based on the response surface results (Fig. 2), warm summers and low snow accumulation. From 1840 to 1900 both larch and hemlock experienced low growth indicating a period of warm summers but high snow accumulations since these conditions result

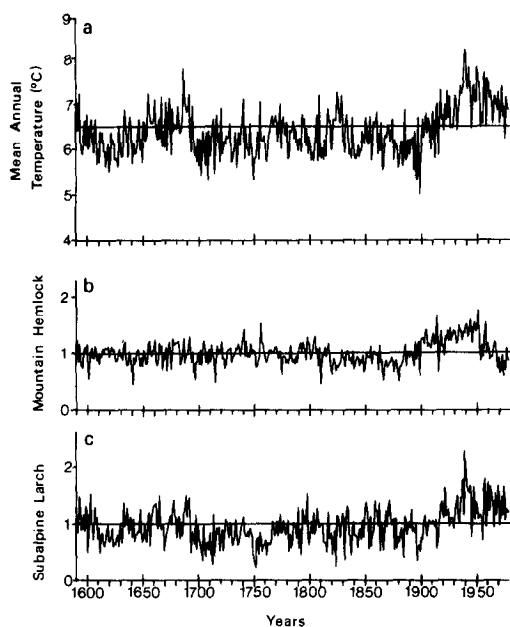


FIG. 5. Reconstruction of mean annual temperature for Longmire, Washington, (a) plotted with the averaged ring width chronologies for all the mountain hemlock (b) and subalpine larch (c) sites.

in the lowest growth rate in both species. This interpretation agrees with Wahl and Lawson's (1970) compilation of historical climate data indicating that during the 1850s and 1860s summer temperatures in Washington were equal to or slightly warmer than temperatures averaged over the period 1931 to 1960, while winter temperatures were more than 0.5°C cooler in the late 19th century as compared to 1931 to 1960. In addition, winter precipitation was more than 10% higher during the 1850s and 1860s. The cooler temperatures and increased precipitation would have resulted in substantially higher snow accumulation at high elevations. The end of the Little Ice Age in the Pacific Northwest thus began with an increase in summer temperatures starting around 1840. Not until 1900, however, did the snow accumulation decrease sufficiently to allow moderate to high growth to occur in these subalpine species.

The period from 1690 to 1760 is reconstructed as a prolonged period of very cool

annual temperatures. This period is a time of very low growth for subalpine larch but average growth for mountain hemlock. Examination of the response surfaces reveals no modern conditions under which this combination of growth patterns occur. Additionally, many of the reconstructed data for this period are identified as extrapolations: over 40% of all the data considered to be extrapolations fall within this 70-yr period. Clearly both the quantitative model and our methods of qualitative inference break down under these conditions. Before the climate of this period can be fully characterized, additional information is needed including long-term growth records from tree species with differing responses to climate as compared to the two species used in this study.

### SUMMARY

Response surface techniques provide an important tool for describing the relationship of tree growth and climate because the growth response of a species to a single climatic variable can change in direction or magnitude depending on the value of a second climatic variable. Response surface results help determine the most biologically reasonable model for quantitatively reconstructing a single climatic variable and also provide the information necessary to assess qualitatively the seasonal nature of past climatic variation.

The growth of subalpine trees in Washington is related to seasonal variation in summer temperature and spring snow depth in a complicated fashion. A model that reconstructs mean annual temperature makes use of the complementary seasonal response of both species to reconstruct past climate. The resulting mean annual temperature reconstruction for Longmire exhibits good correspondence with local proxy records of past temperature but shows little long-term correlation with temperature reconstructions for other regions of the western United States. A notable ex-

ception to this generalization is the coherent rise in temperature indicated by these proxy records at the beginning of the 20th century.

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